Sankarshan Kulkarni

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Professional Summary

Quantitative developer with **2+ years** of experience across **front-office** derivatives pricing and **risk analytics**. Work spans **SABR**, **Black-76**, and **Bachelier** calibration for **SOFR** and **FX** volatility surfaces, and the **Bloomberg-to-kdb+** data infrastructure behind them. Builds in **Python**, **SQL**, **Spark**, **MSSQL**, and **MongoDB**, with production experience on terabyte-scale risk and liquidity reporting pipelines.

Work Experience

Associate (Y3) – PricewaterhouseCoopers Services LLP, India Mar 2026 – Present

Pune, Maharashtra

- Implement and calibrate **SABR**, **Black-76**, and **Bachelier** models to produce **normal (N)**, **lognormal (LN)**, and **shifted-lognormal (SLN)** volatilities across **ATM**, **smile**, and **skew** points for **SOFR swaptions**, **caps/floors**, and **FX options**.
- Handle **negative-rate** and low-strike regimes through shifted-lognormal and normal quoting conventions, including **volatility conversion** across **N/LN/SLN** spaces for consistent cross-desk marks.
- Own the server-side data flow from **Bloomberg (BBG)** into **kdb+/q**: **feed handlers**, **tick capture**, **yield curve** and **volatility surface** persistence, and the **real-time query layer** traders hit intraday for live marks and risk.
- Work directly with **quants** on **model development** and **calibration workflows**, taking research prototypes through **parameter fitting**, **arbitrage-free** and stability checks, and production release on the **pricing library**.
- Build and tune **low-latency** components of the in-house platform using **Python**, **MSSQL**, and **MongoDB**, cutting surface computation and query times on the paths traders depend on during active hours.

Technology Analyst (C10) – Citibank (CSIPL) Jul 2024 – Feb 2026

Pune, Maharashtra

- Built **Python**, **Spark**, and **SQL** pipelines for liquidity risk reporting (**AQUA**) on terabyte-scale datasets feeding **regulatory** and internal risk metrics.
- Developed **feature engineering** and data quality checks across **100+** datasets used for quantitative reporting and downstream model inputs.
- Designed a configuration-driven execution framework for **Spark** and **Impala** workloads with logging, alerting, and monitoring; adopted across **FRMT** risk reporting teams.

Key Projects

BankNifty Dispersion Trade Monitor GitHub

- Developed a real-time dispersion monitor using **WebSocket streaming** (Zerodha KiteTicker) and **multi-threaded API calls** to track net premium across index and constituent straddles, normalised to a **Rs. 6 crore** reference portfolio.
- Achieved **75% reduction** in API calls via smart caching and implemented automatic fallback mechanisms (WebSocket, polling, API, mock) for uninterrupted data delivery.

Short Strangle Backtesting (BankNifty) GitHub

- Built a high-performance 1-min intraday backtest engine processing **10M+** rows of options data; optimised vectorised logic reduced runtime **less than 45 seconds**.
- Computed **Sharpe ratio (1.3)**, **CAGR (8%)**, **max drawdown**, skewness (-1.36), and kurtosis (0.37); validated risk-adjusted performance across 1+ year of tick data.

SVI Volatility Surface Modelling (SPX) GitHub

- Extracted OTM SPXW options (0/1/2 DTE) from a **7.5M-row** dataset; derived forward price using 10-year Treasury yield (r=4.5%) and computed **Black-76 implied volatilities**.
- Fitted a **5-parameter Raw SVI** model via least-squares optimisation, achieving **RMSE 0.00256**; verified no butterfly arbitrage via call price convexity.

Risk Parity & Minimum Volatility Portfolio GitHub

- Implemented two optimisation methods – risk parity (marginal risk contribution) and min-vol with product constraint – on six US stocks using 1-month historical returns.
- Achieved daily variance **10× lower** than market-cap weighted benchmark; risk parity weights evenly distributed (0.12–0.20), confirming true diversification.

Optimised Intraday EMA-RSI Strategy GitHub

- Engineered a **Numba-JIT** accelerated backtesting engine (**5–10x speedup**) with multi-timeframe trend indicators (EMA) and momentum filter (RSI); tuned thresholds via grid search.
- Built an interactive **Streamlit** dashboard providing real-time performance metrics, drawdown analysis, and Sharpe tracking; optimised memory usage for large intraday datasets.

Black-Scholes PDE – Finite Difference GitHub

- Transformed the Black-Scholes PDE to the heat equation and solved it using an **explicit finite difference** scheme on a discretised log-moneyness domain.
- Validated against the analytical Black-Scholes formula, achieving $\mathbf{R^2 = 0.995}$; quantified numerical convergence and stability (CFL condition).

Education

2020 – 2024 B.Tech, Chemical Engineering, **Indian Institute of Technology Gandhinagar**

Technical Skills

Languages & Tools	**Python**, **SQL**, C++, **MSSQL**, MongoDB, **kDB+**, Matlab, Spark, **Pandas**, **NumPy**, **SciPy**, **Statsmodels**, Docker, Git, Linux, Streamlit, Numba, Flask, WebSockets
Quantitative Domains	**Factor Research**, Portfolio Construction, Risk Modeling, Statistical Arbitrage, Time Series Analysis, Monte Carlo Simulation, Derivatives Pricing (Black-Scholes, Black-76), CAPM, Risk Parity, **SVI Volatility Modelling**, Backtesting, Implied Volatility, Finite Difference Methods

Achievements

- Global Rank **87** among **4000+** participants in **NK Securities Quant Research** Challenge for **volatility surface construction** and **option mispricing** identification.
- Winner – Algorithmic Trading Hackathon among **2000+** participants for designing systematic quantitative trading strategies and robust statistical backtesting frameworks.